Textual Analysis in Finance

Week 3: Neural Embedding

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Where are we?

↑ **We are here**

Outline

Word2Vec

Issues with word-count methods

Bag-of-Words (or *tf.idf*)

- ... treats words as **independent units**
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Neural embedding approaches overcome those limitations!!!

The core idea - Embedding

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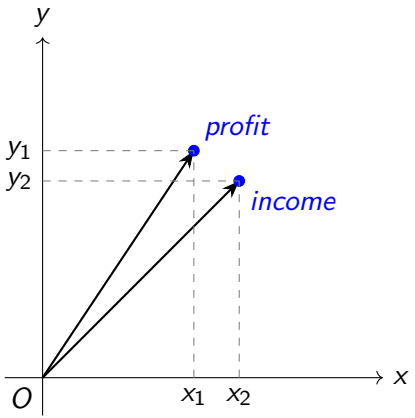

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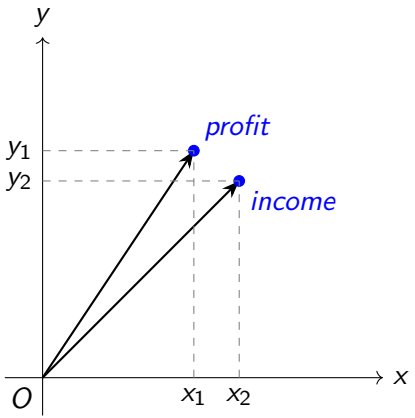

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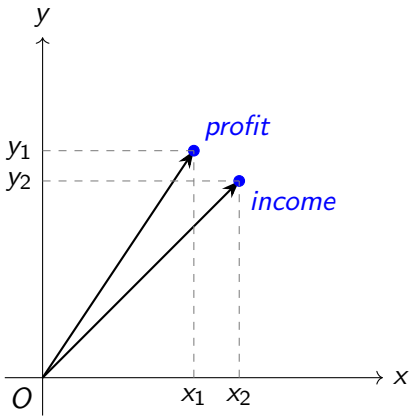

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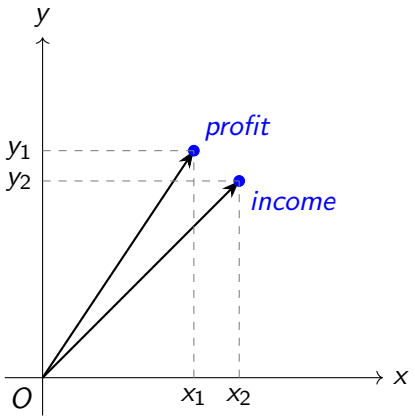

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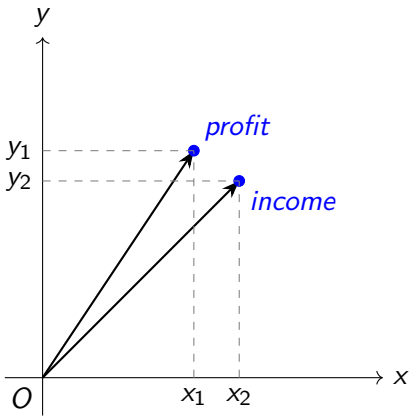

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- We find the embeddings such that ***words that appear in similar contexts should have similar embeddings***

Continuous Bag-of-Words (CBOW) [Mikolov et al., 2013a] - General idea

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- The **embeddings** are the weights from the input layer to the hidden layer

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  - "high" $\rightarrow x_{high} = [0, 0, 1, 0, 0, 0]$
  - "last" $\rightarrow x_{last} = [0, 0, 0, 1, 0, 0]$
  - "quarter" $\rightarrow x_{quarter} = [0, 0, 0, 0, 1, 0]$
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- **Step 4:** The CBOW input is the average of the context word vectors, $x = \frac{1}{4}(x_{reported} + x_{high} + x_{last} + x_{quarter}) = [0, 0, 0.25, 0.25, 0.25, 0.25]$, x is a $1 \times |V|$ vector

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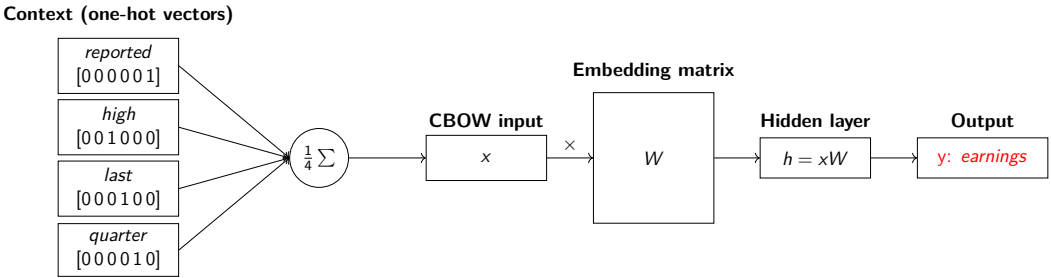

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- **Constructing the CBOW training data.** Start from the text:

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  - Preprocess the text (lowercasing, lemmatization): [“company”, “report”, “high”, “earning”, “last”, “quarter”]
  - Fix a context window size (e.g., window = 2).
  - Slide the window across the text to generate training examples:
    - Target: “company” - Context: {“report”, “high”} - **Edge case, can be ignored**
    - Target: “report” - Context: {“company”, “high”, “earning”}
    - Target: “high” - Context: {“company”, “report”, “earning”, “last”}
    - Target: “earning” - Context: {“report”, “high”, “last”, “quarter”}
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- The predicted output: $\hat{y}_j = \frac{\exp(h\tilde{w}_j^\top)}{\sum_{i=1}^{|V|}\exp(h\tilde{w}_i^\top)}$ ($y = [y_1, ..., y_{|V|}]$; $h = xW$; $\tilde{w}_j$ is a row of $\tilde{W}$) is very costly to compute because of the denominator

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  - Normal CBOW: given a context, predict the correct word among all words in the vocabulary.
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- Example training pairs with negative sampling (context = {"report", "high", "last", "quarter"}):
  - Word: "earning" - Target: 1
  - Word: "company" - Target: 0
  - Word: "loss" - Target: 0
  - Word: "ceo" - Target: 0

  The last 3 examples are called **negative samples** and are randomly chosen!

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- Besides CBOW, there are some other variants of Word2Vec:
  - Skip-gram [Mikolov et al., 2013a]: given a word, predict the context
  - **GloVe** [Pennington et al., 2014]: learns embeddings from a global **co-occurrence matrix**. If words i and j appear together X_{ij} times in the corpus, their embeddings satisfy: $w_i^\top w_j + b_i + b_j \approx \log(X_{ij})$
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- There is Doc2Vec [Le and Mikolov, 2014], which is similar to Word2Vec but for documents (not words)

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- Word2Vec converts words (and documents) into dense vectors, enabling a variety of financial applications:
  - Measure corporate culture from earnings call transcripts [Li et al., 2021] (customized kNN)
  - Identify forward-looking statements in the Management's Discussion and Analysis section of 10-K filings [Brown et al., 2024] (supervised deep learning)
  - Classify business types using Item 1 in 10-K filings [Song, 2021] (kNN)

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- A key challenge of Word2Vec: it relies on labeled data for supervised tasks, which can be limited, making some supervised approaches difficult. E.g., Brown et al. [2024] have to annotate the text themselves

Neural embedding - Pros & Cons

- Pros:
  - Capture semantic meaning: words with similar meanings (e.g., “profit” and “income”) have nearby vector representations
  - Transform sparse, high-dimensional representations (e.g., document–term matrices) into dense, low-dimensional vectors
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- Cons:
  - **Static embeddings** - Once training is complete, each word is mapped to a single vector
    - Context is ignored - "bank" in "commercial bank" and "river bank" share the same vector
    - Poor handling of polysemy; e.g., "charge" can mean a fee or an accusation
  - Limited interpretability - Individual embedding dimensions have no clear semantic or economic meaning
  - Data-hungry - Large corpora are required to learn high-quality embeddings

Contextualized embeddings

Embeddings for Language Model - ELMo [Peters et al., 2018]

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- As a result, ELMo produces different embeddings for "bank" in "commercial bank" and "bank" in "river bank"
- Intuitively: **ELMo = word embeddings + a sequential model**

ELMo's idea

- ELMo treats a sentence as a **time series of words** (w_t):
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- ELMo computes a **contextual embedding** e_t for each word w_t:

$$e_t = f(x_1, \ldots, x_5) \quad \text{or equivalently} \quad e_t = f(x_t, h_t),$$

where $h_t = g(x_1, \ldots, x_{t-1}, x_{t+1}, \ldots, x_5)$ summarizes the surrounding context

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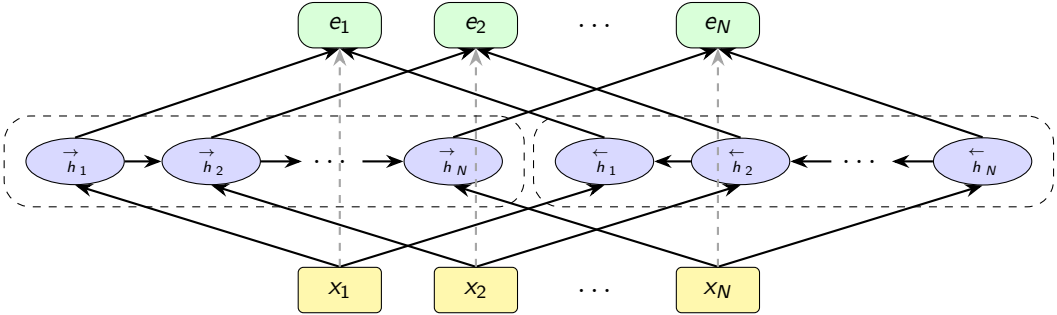
  where $h_t = g(x_1, \ldots, x_{t-1}, x_{t+1}, \ldots, x_5)$ summarizes the surrounding context
- Notice that, $f(x_1, \ldots, x_5)$ also captures the word positions, meaning $f(x_1, x_2, x_3, x_4, x_5) \neq f(x_2, x_1, x_4, x_5, x_3)$

Embeddings for Language Model - ELMo



- $h_t = (\overrightarrow{h}_t^\top, \overleftarrow{h}_t^\top)^\top$, where $\overrightarrow{h}_t = \text{LSTM}(x_t, \overrightarrow{h}_{t-1})$ and $\overleftarrow{h}_t = \text{LSTM}(x_t, \overleftarrow{h}_{t+1})$

  $\rightarrow$ The embedding of a word depends on the words **before** and **after** it in the text!

- E.g., $\overrightarrow{h}_2 = g(x_2, \overrightarrow{h}_1) = g(x_2, g(x_1, \overrightarrow{h}_0))$; similarly $\overleftarrow{h}_{N-1} = g(x_{N-1}, g(x_N, \overleftarrow{h}_{N+1}))$; $\overrightarrow{h}_0, \overleftarrow{h}_N = (0, 0, ..., 0)$

ELMo - Remarks

- ELMo uses **bidirectional Long Short-Term Memory (BiLSTM)** networks [Hochreiter and Schmidhuber, 1997] as its sequential model
- The contextual embedding e_t for word w_t is typically a concatenation of:

$$e_t = \begin{bmatrix} x_t \\ \overrightarrow{h}_t \\ \overleftarrow{h}_t \end{bmatrix} = (x_t^\top, \overrightarrow{h}_t^\top, \overleftarrow{h}_t^\top)^\top$$

- **Context awareness**
  - $\overrightarrow{h}_t$ depends on all previous words $x_1, ..., x_{t-1}$ and hidden states $\overrightarrow{h}_1, ..., \overrightarrow{h}_{t-1}$
  - $\overleftarrow{h}_t$ depends on all future words $x_{t+1}, ..., x_T$ and hidden states $\overleftarrow{h}_{t+1}, ..., \overleftarrow{h}_T$
- The final embedding e_t is therefore **contextual**: the same word w_t gets different embeddings depending on its surrounding words

ELMo - Pros & Cons

- Pros:
  - **Contextualized embeddings**: ELMo addresses key limitations of Bag-of-Words and static embeddings (e.g., Word2Vec) by allowing word representations to vary with context
  - **Improved downstream performance**: consistently outperforms Bag-of-Words and static embeddings in tasks such as sentiment analysis, semantic similarity, and classification

ELMo - Pros & Cons

- Pros:
  - **Contextualized embeddings**: ELMo addresses key limitations of Bag-of-Words and static embeddings (e.g., Word2Vec) by allowing word representations to vary with context
  - **Improved downstream performance**: consistently outperforms Bag-of-Words and static embeddings in tasks such as sentiment analysis, semantic similarity, and classification
- Cons:
  - **Main limitation: reliance on sequential models (LSTMs)**
    - **Computationally expensive** due to sequential processing
    - Difficulty handling very long documents due to **vanishing gradients**
  - **Nearly no applications in finance** - less interpretable than BoW, but less powerful than language models

ELMo is computationally expensive

- Consider a document with T words: $[w_1, w_2, \ldots, w_T]$. The contextualized embedding of the last word w_T is

$$e_T = (x_t^\top, \overrightarrow{h}_t^\top, \overleftarrow{h}_t^\top)^\top$$

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- Due to the sequential nature of the forward LSTM,

$$\overrightarrow{h}_t = g\left(\overrightarrow{h}_{t-1}, x_t\right),$$

which implies

$$\overrightarrow{h}_T = g\left(g\left(\cdots g\left(\overrightarrow{h}_0, x_1\right)\cdots, x_{T-1}\right), x_T\right)$$

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- $\overrightarrow{h}_T$ **cannot be computed in parallel**, so as T grows large:
  - The model must compute $\overrightarrow{h}_1, \overrightarrow{h}_2, \ldots, \overrightarrow{h}_{T-1}$ sequentially
  - This leads to high computational cost and slow training/inference for long documents

ELMo suffers from the vanishing-gradient problem

A neural network is trained by **Stochastic Gradient Descent**, meaning they update the network weights, h, by iteratively subtracting them from the gradients

$$h_t^{(n)} = h_t^{(n-1)} - \eta \frac{\partial L}{\partial h_t}\bigg|_{h_t^{(n-1)}}$$

then, by the chain rule

$$\frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial h_T} \prod_{k=t+1}^{T} \frac{\partial h_k}{\partial h_{k-1}},$$

because, $h_k = g(Wh_{k-1} + b) \rightarrow \frac{\partial h_k}{\partial h_{k-1}} = Wg'(Wh_{k-1} + b)$

$$\left\| \frac{\partial h_k}{\partial h_{k-1}} \right\| \lesssim \|Wg'(.)\|$$

ELMo suffers from the vanishing-gradient problem

Therefore,

$$\left\| \frac{\partial L}{\partial h_t} \right\| \lesssim \|Wg'(.)\|^{T-t}$$

Usually in training, $\|Wg'(.)\| \leq 1$, so $\|Wg'(.)\|^{T-t} \to 0$ as T goes large, or it means

$$\left\| \frac{\partial L}{\partial h_t} \right\| \to 0, \quad T \to \infty$$

- In practice, during SGD, the embeddings of early tokens barely update, i.e., $h_1^{(n)} \approx h_1^{(n-1)}$ across many iterations
- Consequently, ELMo "forgets" ...
  - ... to update the embeddings of early words; and
  - ... long-range dependencies, treating **early words as having almost no influence on the final words** for long documents

Language models solve these problems!

Coming next

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- Hands-on practice
  - Exploring the pre-trained Google Word2Vec model
  - Training Word2Vec and Doc2Vec models using `gensim` and Tesla's annual statements
  - Notebook/Solution
- Next lecture
  - Transformers and Language Models

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